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COLLEGE OF COMPUTER STUDIES

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Chatbot Architecture & Design

1. LLM Model and Access

The chatbot is powered by GROQ AI, utilizing high-speed inference for large language models. Access is provided via API calls, which allows for extremely fast response times and efficient processing of complex queries.

2. System Prompts and Prompting Techniques

The chatbot uses a combination of advanced prompting techniques:
- Role Prompting: The model is instructed to act as a 'powerful code assistant' and a 'senior pair-programmer'.
- Instruction-Following: A detailed set of rules (System Prompts) guides behavior, output style, and tool usage.
- Task Management: The model uses structured tools to plan and track progress.
- Verification-Driven: Instructions mandate verifying changes before final delivery.

3. Chatbot Workflow

Flowchart Description:

```
      [ USER INPUT ]
            |
            v
 [ GATHER CONTEXT ] (Files, Cursor, Linter)
            |
            v
 [ ANALYZE & SEARCH ] (SearchCodebase, Grep)
            |
            v
 [ PLAN TASKS ] (TodoWrite)
            |
            v
 [ EXECUTE CHANGES ] (SearchReplace, Write)
            |
            v
 [ VERIFY & TEST ] (RunCommand, GetDiagnostics)
            |
            v
    [ FINAL RESPONSE ]
```